



Figure 1: Native vs hybrid vs Web



Figure 2: Native vs hybrid vs Web



Figure 3: Webview container

larger compared to, say, Java alone. Hence, a way was found to package the Web application code and run them as apps on mobile devices. This is how hybrid mobile apps have found their way into the industry. Hybrid mobile apps are nothing but HTML5, CSS3 and

JavaScript code wrapped in a webview container and packaged as native apps.

To illustrate the concept of hybrid apps in more depth, let's try and understand how it all actually works. In general, each mobile platform offers a webview container—a UI control/element that helps the user to render HTML5 code inside it. At times, there is a need to show some website or other Web pages within the context of a native app. This is when webview is most useful, and it's this functionality that has opened the doors to hybrid apps. A hybrid app is nothing but a webview control that fits the entire page (Figure 3). So if I were to express this in an 'Android way', it would be something like what follows:

```
<?xml version="1.0" encoding="utf-8"?>
<WebViewxmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/webview"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
/>
```

Those familiar with Android development will be able to make sense of the above code snippet. In Android native development, we usually declare the UI controls and elements in that Activity's XML file. So, for our sample Activity, using the above code, I have created a webview control that fits the entire screen (*fill_parent* for both *layout_height* and *layout_width*). Now, one can write the entire code like it's a Web application and exploit this webview to render the Web page. The Android developers' page has a comprehensive tutorial on how to use webview to make an application: <http://developer.android.com/guide/webapps/webview.html>

In the past few years, a lot of cross-platform tools (CPTs) have emerged for making hybrid apps that let you code in a different language and package your code for Android devices, iOS devices, etc. Most of these tools offer coding languages such as HTML5, CSS3 and JavaScript. Their underlying architecture is almost similar to what we've just discussed. With hybrid apps, the biggest myth is related to performance and accessing device-specific

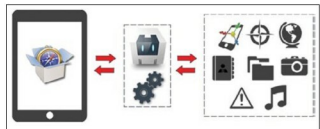


Figure 4: PhoneGap architecture